

Note for section 5

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Abstract

In this note, we discuss the effect of interaction on the topological classification of one-dimensional free fermion with time reversal symmetry $T^2 = 1$. The discussion follows [1–3]

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1 Introduction

Fermion systems are born with an on-site $\mathbb{Z}_2$ symmetry, the fermion parity $(-1)^F$. Therefore, they generally give rise to a richer structure of the topological phases. For example, one-dimensional boson/spin systems, which do not have any symmetry constraint, cannot realize any topological phase. On the contrary, one-dimensional fermion systems can have a non-trivial topological phase characterized by the presence of localized Majorana mode at the edge. If there are two Majorana fermions at each end of the open chain, a local coupling term can easily gap them out. An equivalent viewpoint is to consider two parallel chains. If each has a single Majorana fermion, they can weakly interact and cancel each other. Thus, they fall into a $\mathbb{Z}_2$ classification. A concrete lattice model that realizes both phases is the Kitaev chain

$$H = \frac{1-g}{2} \sum_j i\chi_{2j-1}\chi_{2j} + \frac{1+g}{2} \sum_j i\chi_{2j}\chi_{2j+1} \quad (1)$$

which is in the trivial phase when $g < 0$ and topological phase when $g > 0$. To understand the $\mathbb{Z}_2$ classification in this model, let us consider two parallel Kitaev chain $H = H^{(1)} + H^{(2)}$, both of which are tuned at $g = 1$, the fixed point of the topological phase. Then on the left edge, there are two Majorana zero modes $\chi_1^{(1)}$ and $\chi_1^{(2)}$ giving rise to a two-fold degenerate states. However, a local perturbation $\Delta H = i\chi_1^{(1)}\chi_1^{(2)}$ can easily lift the degeneracy and trivialize the system. Similar analysis applies to the right edge.

Now, let us add one more symmetry to the system, the time-reversal symmetry T . Since we are considering spinless fermions, the time-reversal is defined by¹

$$TcT^{-1} = c, \quad T^2 = 1, \quad T(-1)^F = (-1)^FT \quad (2)$$

where c is the annihilation operator of the complex fermions. Thus, the time-reversal acts on the Majorana fermion in the following way

$$T\chi_{2j-1}T^{-1} = \chi_{2j-1}, \quad T\chi_{2j}T^{-1} = -\chi_{2j} \quad (3)$$

This additional symmetry constrains the allowed local perturbation and enlarges the space of topological phases. If we disallow interaction, namely, the Hamiltonian must be a fermion bilinear, the topological classification is augmented from $\mathbb{Z}_2$ to $\mathbb{Z}$. For example, we can consider N parallel Kitaev chains at $g = 1$. All the quadratic coupling terms between the localized Majorana modes on the same edge violates the time-reversal, e.g.

$$T \left(i \sum_{jk} A_{jk} \chi_1^{(j)} \chi_1^{(k)} \right) T^{-1} = - \left(i \sum_{jk} A_{jk} \chi_1^{(j)} \chi_1^{(k)} \right)$$

¹The time-reversal symmetry in real-world fermion systems always satisfies $T^2 = (-1)^F$ rather than the one $T^2 = 1$ that we are considering here. To make a connection between them, we can regard the time-reversal T in this note to be related to the actual one T_{actual} concatenated with a reflection, i.e. $T = RT_{actual}$ such that it preserves the direction of spin and squares to 1. More explicitly, let R be the reflection with respect to the xy plane, then we have

$$T = RT_{actual} = \sigma_x i\sigma_y K = -\sigma_z K$$

It acts diagonal in the σ_z basis and squares to 1.

and are forbidden. Therefore, all the N Majorana modes can survive, which gives rise to the $\mathbb{Z}$ classification.

However, the story can change dramatically when interactions, quartic terms of the fermions, are allowed. There are two possible scenarios:

- Some inequivalent topological phases now become equivalent in the presence of interaction. Namely, one can find a smooth path in the space of Hamiltonian that connects the two phases without going through a transition.
- New topological phases can arise, which do not have any free-fermion analogue. [Maybe the inverse problem is interesting, i.e., given an anomaly, whether it is realizable in free fermion.]

It turns out that only the first scenario happens in this case and the classification is reduced from $\mathbb{Z}$ to $\mathbb{Z}_8$. The eight different phases can be constructed from N parallel Kitaev chains with symmetric interaction, and they are physically characterized by either the projective representation or unpaired Majorana mode on the edge for N being even or odd, respectively. We will call them a mod-8 anomaly on the edge. The results are summarized below

$N \bmod 8$	T^2	$[T, (-1)^F]_\zeta = 0$	$N \bmod 8$	T^2
0	1	1	1	1
2	1	-1	3	-1
4	-1	1	5	-1
6	-1	-1	7	1

The data for $N \bmod 8$ being odd requires some clarification and we will come to that in the later section.

In the rest of this note, we will explicitly compute the mod 8 anomaly and thus the $\mathbb{Z}_8$ classification. We will also generalize our discussion to the case $T^2 = (-1)^F$ and show that it exhibits a mod-2 anomaly. To further confirm the classification, we will explicitly construct a path in the Hamiltonian space that smoothly connects the eight Kitaev chains in the trivial phase and in the (free-fermion) topological phase.

2 Mod-8 anomaly

In this section, we will explicitly compute the anomaly on the edge to determine the classification. Our strategy is the following. We consider N parallel Kitaev chains, each of which is prepared in the (free-fermion) topological phase. There are N localized Majorana modes $\lambda_{i=1,2,\dots,N}$ on the left edge and another N Majorana modes $\tilde{\lambda}_{i=1,2,\dots,N}$ on the right edge. They transform in the opposite way under the time-reversal

$$T\lambda T^{-1} = \lambda, \quad T\tilde{\lambda} T^{-1} = -\tilde{\lambda}$$

The reason that we use λ instead of χ is to emphasize that the Kitaev chains are not necessarily at the $g = 1$ fixed point. Without loss of generality, we can focus on the left edge and abstract

the question as follows. The system has N Majorana fermions λ_i and a symmetry $\mathbb{Z}_2 \times \mathbb{Z}_2$. One is the fermion parity $(-1)^F$ and the other is the time-reversal T . The two symmetries, when acting on the elementary fermions, commute with each other and obey $T^2 = 1$. When acting on the N Majorana λ_i 's, they are only required to satisfy the following more basic algebraic relation

$$\{\lambda_i, \lambda_j\} = 2\delta_{ij}, \quad \{(-1)^F, \lambda_i\} = 0, \quad T\lambda_i T^{-1} = \lambda_i \quad (4)$$

We want to know whether the system is anomalous. When it is so, there does not exist any local perturbation that can trivialize the system.

2.1 N is even

In this case, the Majorana fermions form a Hilbert space of dimension $2^{N/2}$. Therefore, we can try to construct all the operators explicitly as certain matrices acting on the Hilbert space that satisfy the desired algebraic relations listed in (4). The anomaly, which is synonymous to having a projective representation here, will manifest then.

N = 2 In this case, there is no quartic interaction and thus the two-fold degeneracy cannot be lifted. Now, let us understand it from the perspective of projective representation. Given the the algebraic relations to satisfy, we can represent the operators by the following 2×2 matrices

$$\lambda_1 = \sigma^x, \quad \lambda_2 = \sigma^y, \quad (-1)^F = \sigma^z, \quad T = \sigma^x K \quad (5)$$

We can find that the time-reversal symmetry *anti-commutes* with the fermion parity

$$T(-1)^F = -(-1)^F T \quad (6)$$

There must be at least a two-fold degeneracy in order to support this anti-commutator. This is the algebraic reason that why the two-fold degeneracy cannot be removed.

N = 4 The matrix representation of the operators can be constructed by re-cycling what we have had already. Let $\hat{\lambda}_1, \hat{\lambda}_2, (-1)^{\hat{F}}$ and $\hat{T}$ denote the operators we have built for the $N = 2$ case. Then, the operators for the $N = 4$ case can be written as²

$$\begin{aligned} \lambda_1 &= \hat{\lambda}_1 \otimes \mathbb{I}, \quad \lambda_2 = \hat{\lambda}_2 \otimes \mathbb{I}, \quad \lambda_3 = (-1)^{\hat{F}} \otimes \hat{\lambda}_1, \quad \lambda_4 = (-1)^{\hat{F}} \otimes \hat{\lambda}_2 \\ (-1)^F &= (-1)^{\hat{F}} \otimes (-1)^{\hat{F}}, \quad T = \hat{T} \otimes \left(\hat{T}(-1)^{\hat{F}} \right) \end{aligned} \quad (7)$$

It becomes clear that T still commutes with $(-1)^F$ but

$$T^2 = \hat{T}^2 \otimes \left(\hat{T}(-1)^{\hat{F}} \hat{T}(-1)^{\hat{F}} \right) = -1 \quad (8)$$

This implies that there must be at least a two-fold Kramer degeneracy, that cannot be removed by any perturbation.

²This is nothing but the Jordan-Wigner transformation.

An equivalent way of extracting this $T^2 = -1$ projective representation is to construct the states in the Hilbert space and examine how T and $(-1)^F$ acts on them explicitly. For example, we can define two complex fermions c_1, c_2 by

$$c_1 = \frac{\lambda_1 + i\lambda_2}{2}, \quad c_2 = \frac{\lambda_3 + i\lambda_4}{2} \quad (9)$$

and use the Fock space to construct the Hilbert space. Let $|00\rangle$ denote the vacuum of c_1 and c_2 , the Hilbert space is

$$|00\rangle, \quad |11\rangle \equiv c_2^\dagger c_1^\dagger, \quad |01\rangle \equiv c_1^\dagger |00\rangle, \quad |10\rangle \equiv c_2^\dagger |00\rangle \quad (10)$$

We emphasize that the ordering of $c^\dagger$'s in the definition is important because the $\pm$ signs are exactly what we are interested in here. Because $Tc_iT^{-1} = c_i^\dagger$, T can act within the two subspaces $\{|00\rangle, |11\rangle\}$ and $\{|10\rangle, |01\rangle\}$, e.g. $T|00\rangle = e^{i\phi}|11\rangle$, $T|01\rangle = e^{i\theta}|10\rangle$, where ϕ and θ are unimportant phase factors. One can check how T^2 acts on either of these four states and verify $T^2 = -1$.

In this case, there is indeed one T -invariant quartic interaction $\lambda_1\lambda_2\lambda_3\lambda_4$. Its effect is to select one of the two-dimensional subspace as the ground states but cannot lift the degeneracy between that two states.

Let us give one more remark before moving onto the next case. We can see that the projective representation is given only by the time-reversal and has nothing to do with the fermion parity. Therefore, this case is actually a bosonic SPT. It is the famous Haldane phase (AKLT phase, cluster state). Reversely, it means that the two-fold edge degeneracy of four copies of Kitaev chains can be regarded as a spin-1/2 degree of freedom.

N = 6 Again, let us construct the operators based on what we have had already. Let $\hat{\lambda}_1, \hat{\lambda}_2, (-1)^{\hat{F}}$ and $\hat{T}$ denote the operators for the $N = 2$ case. Let $\lambda'_{i=1,2,3,4}, (-1)^{F'}$ and T' denote the operators for the $N = 4$ case. Then, we have

$$\begin{aligned} \lambda_1 &= \hat{\lambda}_1 \otimes \mathbb{I}, \quad \lambda_2 = \hat{\lambda}_2 \otimes \mathbb{I}, \quad \lambda_3 = (-1)^{\hat{F}} \otimes \lambda'_1, \\ \lambda_4 &= (-1)^{\hat{F}} \otimes \lambda'_2, \quad \lambda_5 = (-1)^{\hat{F}} \otimes \lambda'_3, \quad \lambda_6 = (-1)^{\hat{F}} \otimes \lambda'_4, \\ (-1)^F &= (-1)^{\hat{F}} \otimes (-1)^{F'}, \quad T = \hat{T} \otimes (T'(-1)^{F'}) \end{aligned} \quad (11)$$

It is then clear that

$$\begin{aligned} (-1)^F T &= \left((-1)^{\hat{F}} \hat{T} \right) \otimes \left((-1)^{F'} T' (-1)^{F'} \right) = -T (-1)^F \\ T^2 &= \left(\hat{T}^2 \right) \otimes \left(T' (-1)^{F'} T' (-1)^{F'} \right) = -1 \end{aligned} \quad (12)$$

Therefore there is at least a two-fold degeneracy that cannot be removed by any symmetry allowed perturbation.

N = 8 We construct the operators by taking the tensor product of two $N = 4$ cases. Let $\lambda'_{i=1,2,3,4}$, $(-1)^{F'}$ and T' denote the operators for the $N = 4$ case. Then, we have

$$\begin{aligned} \lambda_1 &= \lambda'_1 \otimes \mathbb{I}, & \lambda_2 &= \lambda'_2 \otimes \mathbb{I}, & \lambda_3 &= \lambda'_3 \otimes \lambda'_1, & \lambda_4 &= \lambda'_4 \otimes \lambda'_2, \\ \lambda_5 &= (-1)^{F'} \otimes \lambda'_1, & \lambda_6 &= (-1)^{F'} \otimes \lambda'_2, & \lambda_7 &= (-1)^{F'} \otimes \lambda'_3, & \lambda_8 &= (-1)^{F'} \otimes \lambda'_4 \\ & & (-1)^F &= (-1)^{F'} \otimes (-1)^{F'}, & T &= T' \otimes T' \end{aligned} \quad (13)$$

The fermion parity and time-reversal return to the regular representation and this case becomes completely trivial.

Readers might worry that whether there could be a more subtle anomaly that we have not found but actually exists when N is a multiple 8. This is in principle possible but does not happen here. The intuition for this is to recall the identification of $N = 4$ with the Haldane phase, the later has a $\mathbb{Z}_2$ classification. We can also explicitly write down a Hamiltonian that has a unique ground state. The edge of $N = 4$ can be regarded as a spin-1/2. Thus, the edge of $N = 8$, if it had two spin-1/2 degrees of freedom, can be easily gapped out by an anti-ferromagnetic Heisenberg coupling. Since anomalies can always be captured by any description of the system that is valid at low energies, this proves that anomaly cannot exist in this case.

2.2 N is odd

The case that N is odd is a bit more subtle because we need an even number of fermions to have a locally well-defined Hilbert space. So the anomaly is not (at least not only) associated to having a projective representation but the unpaired Majorana fermions.

Using the Majorana fermion operators, we can write down an operator, which is simply the product of all of them

$$\lambda_1 \lambda_2 \cdots \lambda_N \quad (14)$$

that *commutes* with all the λ_i and thus it can be chosen as a complex number. Note that $(\lambda_1 \cdots \lambda_N)^2 = (-1)^{(N-1)N/2} = (-1)^{(N-1)/2}$, there are two choices

$$\lambda_1 \lambda_2 \cdots \lambda_N = \pm i^{(N-1)/2} \quad (15)$$

Pick one and we obtain a Hilbert space of dimension $2^{(N-1)/2}$. The fermion parity $\lambda_i \rightarrow -\lambda_i$ flips the sign of this c -number and does not act within each of them. Such a structure cannot be lifted by any Hamiltonian because a Hamiltonian always involves an even number of fermions.

Physically, the above procedure is exactly related to the existence of unpaired Majorana modes. It suffices to explain in the simplest case $N = 1$, where there is λ and $\tilde{\lambda}$ on the two edges respectively. We can combine them into a complex fermion $c = \frac{1}{2}(\lambda + i\tilde{\lambda})$ and two ground states correspond to the two Fock states $|0\rangle$ and $|1\rangle$ of c . Choosing $\lambda = \pm 1$ amounts to choosing the ground state to be $\frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$. Therefore, more generally, picking one choice in (15) means that we want to work with a particular ground state of the whole system, and any Hamiltonian cannot alter the choice means that the degeneracy of the ground state is robust.

Let us now explain the “not only” part of the statement at the beginning of this subsection. After we make the choice in (15) and obtain the $2^{(N-1)/2}$ -dimensional Hilbert space, let us

inquire how the symmetries act and how we can further distinguish the four case with an odd N . We can define that a new time-reversal operator $\mathcal{T}$ as

$$\mathcal{T} = \lambda_1 \lambda_3 \cdots \lambda_N K \quad (16)$$

which obeys

$$\mathcal{T} \lambda_i \mathcal{T}^{-1} = \begin{cases} \lambda_i & N = 1, 5 \pmod{8} \\ -\lambda_i & N = 3, 7 \pmod{8} \end{cases}, \quad \mathcal{T}^2 = \begin{cases} 1 & N = 1, 7 \pmod{8} \\ - & N = 3, 5 \pmod{8} \end{cases} \quad (17)$$

Therefore, we can regard it as the ordinary T when $N = 1, 5 \pmod{8}$ or the product of T and $(-1)^F$ when $N = 3, 7 \pmod{8}$, and it can have a projective representation.

Remark Let us close the whole discussion on the mod-8 anomaly by a small generalization and a summary. So far we have only included the Majorana fermions λ_i that are even under the time-reversal in our system. This is natural when we consider N parallel Kitaev chains. However, Majorana fermions $\tilde{\lambda}_i$ that are odd under the time-reversal should also be allowed in a more general setup. They can be used to gap out the λ fermion as we have mentioned before.

Therefore, the mod-8 anomaly in the most general case can be summarized as follows. There are two kinds of fermions $\lambda_i, i = 1, \dots, r$ and $\tilde{\lambda}_i, i = 1, \dots, s$. They are in the linear representation of our $\mathbb{Z}_2^F \times \mathbb{Z}_2^T$ symmetry. i.e.

$$\{(-1)^F, \lambda_i\} = \{(-1)^F, \tilde{\lambda}_i\} = 0, \quad T \lambda_i T^{-1} = \lambda_i, \quad T \tilde{\lambda}_i T^{-1} = -\tilde{\lambda}_i \quad (18)$$

There is a mod-8 anomaly depending on $(r - s) \pmod{8}$. This result was first obtained by mathematician Atiyah, Bott and Shapiro in 1964.

2.3 Remark on the case $T^2 = (-1)^F$

Our discussion can be generalized to the case that $T^2 = (-1)^F$.³ We will show that it has a mod-2 anomaly. Namely, one-dimensional fermion systems that have the time reversal $T^2 = (-1)^F$ fall into a $\mathbb{Z}_2$ classification.

This case is simpler than the previous one with $T^2 = 1$ because the number of fermions must be even. Let us consider N Majorana fermions and divide them into two groups $\lambda_i, i = 1, \dots, N/2$ and $\tilde{\lambda}_i, i = 1, \dots, N/2$. They form a $2^{N/2}$ -dimensional Hilbert space with the fermion parity $(-1)^F = i^{N/2} \chi_1 \tilde{\chi}_1 \cdots \chi_{N/2} \tilde{\chi}_{N/2}$. They can be represented by certain matrices in a straightforward way using the Jordan-Wigner transformation. Our remaining task is to find a representation for the time-reversal transformation that satisfies $T^2 = (-1)^F$. One can check a few simple case $N = 2$ or 4 and find the following one

$$T = \frac{\chi_1 + \tilde{\chi}_1}{\sqrt{2}} \cdots \frac{\chi_{N/2} + \tilde{\chi}_{N/2}}{\sqrt{2}} K \quad (19)$$

³So $T^4 = 1$ and T generates a group $\mathbb{Z}_4^T$. It also implies $[T, (-1)^F] = 0$ when acting on the fundamental fermions.

It flips the two groups λ and $\tilde{\lambda}$ as

$$T\lambda_i T^{-1} = \tilde{\lambda}_i, \quad T\tilde{\lambda}_i T^{-1} = -\lambda_i$$

So-defined T involves $N/2$ fermions operator and thus commute with $(-1)^F$ when $N = 0 \bmod 4$ but *anti-commute* with $(-1)^F$ when $N = 2 \bmod 4$. Equivalently, we can find that

$$T^2 = i^{N^2/4} (-1)^F \quad (20)$$

If $N = 2 \bmod 4$, we have $T^2 = i(-1)^F$. If we absorb the factor of i into the definition of $(-1)^F$, we would get $((-1)^F)^2 = -1$. So, *we can move the anomaly around, but there is no way to avoid it.*

Thus we have found an anomaly that depends on the value of $N \bmod 4$. There is no more subtle anomaly, since there are symmetric perturbation that can remove all pairs of fermions from the low energy spectrum. For example, when $N = 2$, we can choose $\Delta H = im(\lambda_1 \lambda_2 + \tilde{\lambda}_1 \tilde{\lambda}_2)$. Recall that N is even, this is a mod-2 anomaly.

3 Adiabatic continuation

In this section, we explicitly construct a smooth path that connects eight decoupled Kitaev chains that are in the trivial phase to eight decoupled chain that are in the (free-fermion) topological phase. This further confirms the $\mathbb{Z}_8$ classification.

It suffices to showing that some representative points in the two phases can be smoothly connected. We will pick the most convenient ones, $g = 1$ and $g = -1$. Our strategy is the following:

1. We consider 8 parallel Kitaev chains at the fixed point $g = 1$ of the trivial phase, i.e., $H_0 = \sum_{\alpha=1}^8 H^{(\alpha)}$ and $H^{(\alpha)}$ only contains intra-cell couplings

$$H^{(\alpha)} = i \sum_j \chi_{2j-1}^{(\alpha)} \chi_{2j}^{(\alpha)}$$

So we do not need to work with a long chain but can focus on the finite-dimensional subspace spanned by $\chi_{2j-1}^{(\alpha)}, \chi_{2j}^{(\alpha)}, \alpha = 1, 2, \dots, 8$.

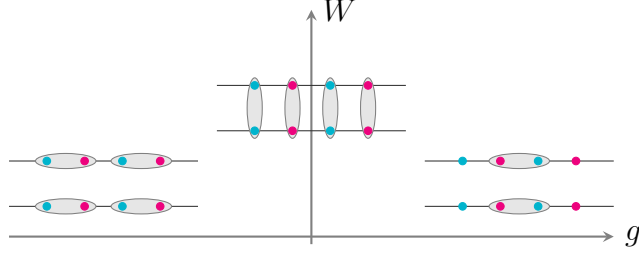
2. Turn on appropriate quartic couplings W , which only involves products of Majorana fermions with the *same* site index, i.e.

$$W \supset \chi_j^{(\alpha_1)} \chi_j^{(\alpha_2)} \chi_j^{(\alpha_3)} \chi_j^{(\alpha_4)}$$

When the interaction is tuned to be strong enough, we turn off the intra-cell coupling.

3. Then, turn on the inter-cell coupling $i\chi_{2j}^{(\alpha)} \chi_{2j-1}^{(\alpha)}$, turn off the interaction. This leads to 8 parallel Kitaev chains at the fixed point $g = -1$ of the (free-fermion) topological phase,

The procedure can schematically shown as



Note the the third step is symmetric with respect to the first two steps, we just need to interchange the even and odd sites. Therefore, we only have to make sure that the interaction can keep the system gapped in the first two steps.

3.1 Majorana fermions and representation of $so(N)$

The choice of interaction term is motivated by the identification of the $N = 4$ case with the Haldane chain. In order to explain the details, we have to first review some relevant mathematics.

In general, $2n$ Majorana fermions $\lambda_1, \lambda_2, \dots, \lambda_{2n}$ form a 2^n -dimensional Hilbert space $\mathcal{H}$. Now, we show that $\mathcal{H}$ is a representation of the Lie algebra $so(2n)$. Given a skew-symmetric matrix $A \in so(2n)$, we define ⁴

$$\rho(iA) = \frac{i}{4} \sum_{j,k=1}^{2n} A_{jk} \lambda_j \lambda_k \quad (21)$$

where i is to make the operator Hermitian and to obtain a unitary representation. We can check that

$$[\rho(iA), \rho(iB)] = i\rho(i[A, B]) \quad (22)$$

so that ρ defines a map (representation) of Lie algebras. The induced action on λ is

$$\lambda_j \rightarrow \lambda'_j = i[\rho(iA), \lambda_j] = \sum_{jk} A_{jk} \lambda_k \quad (23)$$

Let us look at one example $so(4) = so(3) \oplus so(3)$, which will be helpful for our construction of the Hamiltonian. $so(4)$ has six generators. Geometrically, they generate the rotation in the three-dimensional subspace (x_1, x_2, x_3) and the mixture between the three axis and the fourth direction x_4 . In the fundamental representation, we can write the six generators as⁵

$$(J_{(mn)})^{jk} = -(\delta^{mj} \delta^{nk} - \delta^{mk} \delta^{nj}) \quad (24)$$

⁴Note that our convention is different from that in [1].

⁵Sometimes, physicists also add a factor of i here, but we will not do it here in order be consistent with the convention in the general discussion above.

We can make superposition of them to get the direct sum of two $so(3)$ as follows

$$\begin{aligned} J_{\pm,1} &= \frac{1}{2}(J_{23} \pm J_{14}) \\ J_{\pm,2} &= \frac{1}{2}(J_{31} \pm J_{24}) \\ J_{\pm,3} &= \frac{1}{2}(J_{12} \pm J_{34}) \end{aligned} \quad \Rightarrow \quad \begin{aligned} [J_{\pm,i}, J_{\pm,j}] &= \epsilon_{ijk} J_{\pm,k} \\ [J_{\pm,i}, J_{\mp,j}] &= 0 \end{aligned} \quad (25)$$

Accordingly, we can use 4 Majorana fermions (basically the edge of the $N = 4$ case) to represent the above generators by

$$\begin{aligned} \rho(iJ_{\pm,1}) &= -\frac{i}{4}(\lambda_2\lambda_3 \pm \lambda_1\lambda_4) \\ \rho(iJ_{\pm,2}) &= -\frac{i}{4}(\lambda_3\lambda_1 \pm \lambda_2\lambda_4) \\ \rho(iJ_{\pm,3}) &= -\frac{i}{4}(\lambda_1\lambda_2 \pm \lambda_3\lambda_4) \end{aligned} \quad \Rightarrow \quad \begin{aligned} [\rho(iJ_{\pm,i}), \rho(iJ_{\pm,j})] &= i\epsilon_{ijk}\rho(iJ_{\pm,k}) \\ [\rho(iJ_{\pm,i}), \rho(iJ_{\mp,j})] &= 0 \end{aligned} \quad (26)$$

The quadratic Casimir has a simple expression in terms of the four Majorana fermions

$$J_{\pm}^2 = \frac{3}{8}(1 \mp \lambda_1\lambda_2\lambda_3\lambda_4) \quad (27)$$

where $\lambda_1\lambda_2\lambda_3\lambda_4$ is exactly the only quartic interaction that we can write down for the four Majorana fermions. Therefore, the Hilbert space can be decomposed into a direct sum of two spin-1/2 representations, i.e.

$$\mathcal{H} = (0, \frac{1}{2}) \oplus (\frac{1}{2}, 0) \quad (28)$$

For example, the two states $|\psi\rangle$, which satisfy $\lambda_1\lambda_2\lambda_3\lambda_4|\psi\rangle = |\psi\rangle$, are annihilated by J_+ and form a spin-1/2 representation of J_- because $J_-^2|\psi\rangle = \frac{3}{4}|\psi\rangle$. Similarly, the other two states $|\phi\rangle$, satisfying $\lambda_1\lambda_2\lambda_3\lambda_4|\phi\rangle = -|\phi\rangle$, form a spin-1/2 representation of J_+ .

3.2 Construction of the smooth path

Now, we are ready to explain what interaction terms to choose. Let us divide the 16 Majorana fermions into the following four groups

$$\begin{aligned} A &: \{\chi_{2j-1}^{(1)}, \chi_{2j-1}^{(2)}, \chi_{2j-1}^{(3)}, \chi_{2j-1}^{(4)}\}, & B &: \{\chi_{2j}^{(1)}, \chi_{2j}^{(2)}, \chi_{2j}^{(3)}, \chi_{2j}^{(4)}\}, \\ C &: \{\chi_{2j-1}^{(5)}, \chi_{2j-1}^{(6)}, \chi_{2j-1}^{(7)}, \chi_{2j-1}^{(8)}\}, & D &: \{\chi_{2j}^{(5)}, \chi_{2j}^{(6)}, \chi_{2j}^{(7)}, \chi_{2j}^{(8)}\} \end{aligned}$$

Within each group, we first turn on the quartic coupling

$$W_1 = \sum_{k=2j-1, 2j} \chi_k^{(1)} \chi_k^{(2)} \chi_k^{(3)} \chi_k^{(4)} + \chi_k^{(5)} \chi_k^{(6)} \chi_k^{(7)} \chi_k^{(8)}$$

to get the two-fold degenerate ground states. And each two degenerate ground states form a spin-1/2 representation of $so(3)_+$. Then, we turn on an anti-ferromagnetic coupling between different groups

$$W_2 = \sum_{i=1}^3 \rho(iJ_{+,i})^A \rho(iJ_{+,i})^C + \rho(iJ_{+,i})^B \rho(iJ_{+,i})^D$$

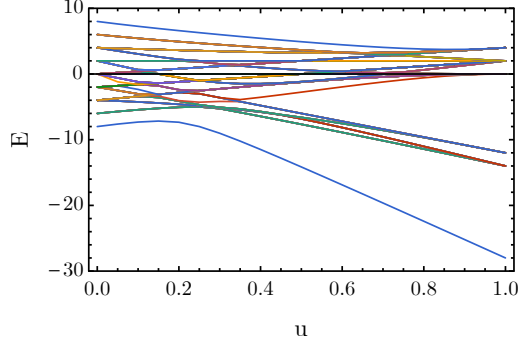


Figure 1: The spectrum of the 16 Majorana fermion in the adiabatic continuation. The ground state is always separated from the excited by a nonzero gap.

This will further gap out the four spin-1/2's and give a unique ground state that contains two the singlet states. The full interaction term is

$$W = W_1 + 16W_2 \quad (29)$$

where the factor of 16 is not necessary but just to make the formula more uniform when it is written in terms of χ 's. The smooth path that can does the step 1 and 2 is

$$H(u) = (1 - u)H_0 + uW, \quad u \in [0, 1] \quad (30)$$

As u goes from 0 to 1, the kinetic term is gradually turned off and the interaction term dominates. We can numerically check that the spectrum remains gapped in the entire process and the result is shown in Fig. 1.

References

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